



Liquid fuels demand and supply model

Week 1 analytical pack

South Africa | Petrol and diesel

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



Vopak's 2026 Liquid Fuels Outlook shows sustained import volumes, with growth from accessible customers (cost × service)

Agenda	SCR	What the evidence shows	So what / story verdict	Deep dives
Market baseline	S1	Diesel drives imports; sales reports conflict 2024 net petrol/diesel imports: 13.078 bn L. Production, stocks, sales coverage and port allocation remain open.	Supported: imports are material. Sustained future volumes remain a forecast case; the matched balance is open.	3 / 6
	S2	Demand sits around Vopak's footprint GP, KZN and WC account for 68.6% of 2022 sales. Historical volumes were broadly steady; later provinces are estimates.	Supported: concentrated demand. Focus on Gauteng/KZN; sales totals do not establish accessible customers.	4 / 5
Market changes	S/C3	Demand drivers differ; L/M/H cases cover all key levers Power, freight, fleet, prices, supply and customer access need explicit assumptions; fuel effects remain uncalibrated.	Partly supported: diesel generation has fallen. Baseline, ample and tight worlds still need sourced assumptions and tested fuel/customer-volume effects.	7 / 9
	C4	Supply change makes import needs conditional Capacity has declined; redevelopment is conditional. Secunda/Natref product output and future cases remain open.	Supported: refining capacity fell. Plant futures are conditional; capacity alone does not determine imports.	8 / 9
	C5	Alternative gateways challenge the same inland market Matola and Walvis Bay have storage; their competitiveness depends on full costs, route capacity and access.	Open: gateway competition. No route winner without comparable cost, service and customer access.	10 / 11
	R6	Vopak footprint is established; turnover estimates handling Page 13 tests 1 / 2 / 3 monthly turns on the published inventory. Actual unique deliveries remain unquantified.	Estimate handling now; validate customer share. Gross turnover equivalents need usable space, dispatch limits and customer access.	12 / 13 / 14
Vopak outlook	R7	Potential scope for new investment? Prioritise Durban/Lesedi customer flows. Additional storage follows secured volumes and a usable-capacity gap.	Conditional: win customers first. Use existing assets; new storage needs secured flows and a usable-capacity gap.	15 / 16 / 19

Note: S = Situation C = Complication R = Resolution | Every Henry storyboard prompt has a response on pages 20–22; open items remain open.

Source: Partner SA storyboard; SARS / Stats SA / operator evidence, 6 Oct 2026. Remaining gaps labelled.

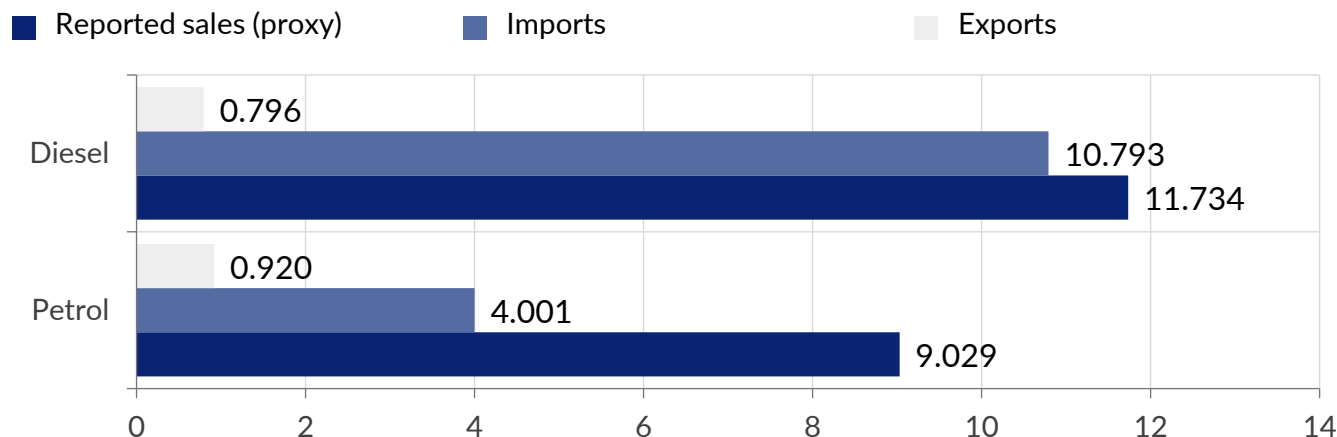
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S1. Diesel drives import volumes, while conflicting reports leave 2024 sales unresolved

National petrol/diesel | 2024, billion litres

Production + imports – exports – stock build = consumption

Reported sales are a consumption proxy; production and stock changes are missing.



Net imports: petrol 3.081, diesel 9.997; combined 13.078 bn litres.

Primary trade | SARS customs

Diesel imports 10.793 bn L.

Government rounds to 10.800; FIASA 14.793.

Use SARS; retain the source discrepancy.

2024 accounting remains open

Sales less net imports = 7.685 bn L.

Balancing requirement, not production.

Production, stocks and coverage unresolved.

What the evidence shows

01 | Customs supports the trade selection

SARS records diesel imports of 10.793 and exports of 0.796 bn litres in 2024. Net imports are 9.997; use customs as the primary trade source.

02 | Reported sales remain a proxy

FIASA reports petrol/diesel sales of 20.763 bn litres for 2024. Editions differ; this demand proxy remains unverified and is not imports plus exports.

03 | Production and stocks remain open

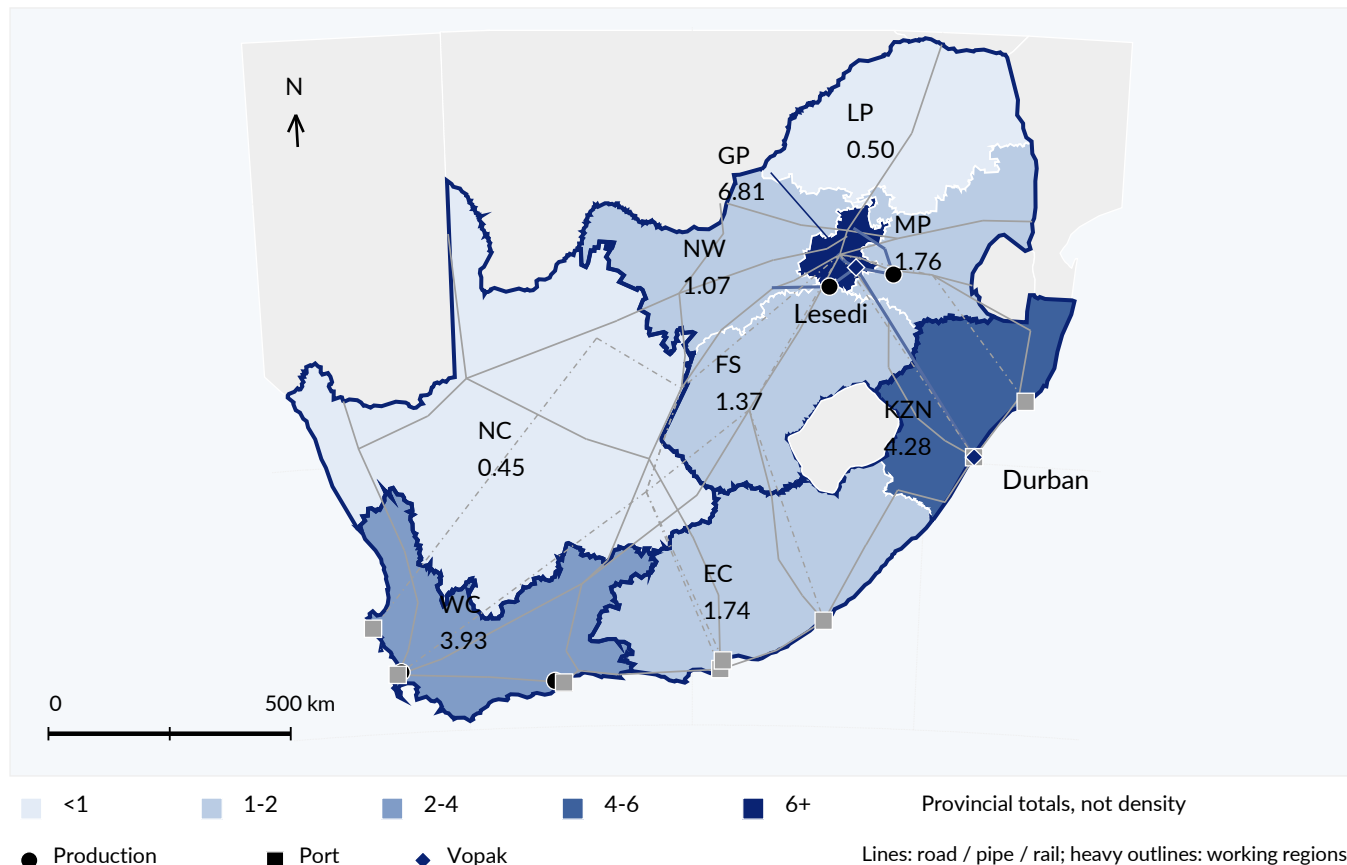
Production and stocks remain unresolved. The proposed 3-4 bn litre diesel gap is a hypothesis, not measured consumption.

To resolve next

Manish: rebuild the balance CSV from SARS; match product, year, actual output and stock changes. Nigel: agree the sales selection and obtain customer/port evidence.

S2. Three provinces account for 69% of reported fuel sales; the top two contain Vopak's storage facilities

Petrol + diesel sales by province | 2022, bn litres/year



What the evidence shows

01 | Demand is concentrated

Gauteng, KwaZulu-Natal and Western Cape account for 68.6% of reported 2022 petrol/diesel sales.

02 | Four reporting regions

Eastern coast: EC/KZN. Inland: GP/FS/LP/MP/NW. Western coast: WC. Other: NC. Boundaries are reporting groups.

03 | Sales locate the historical market

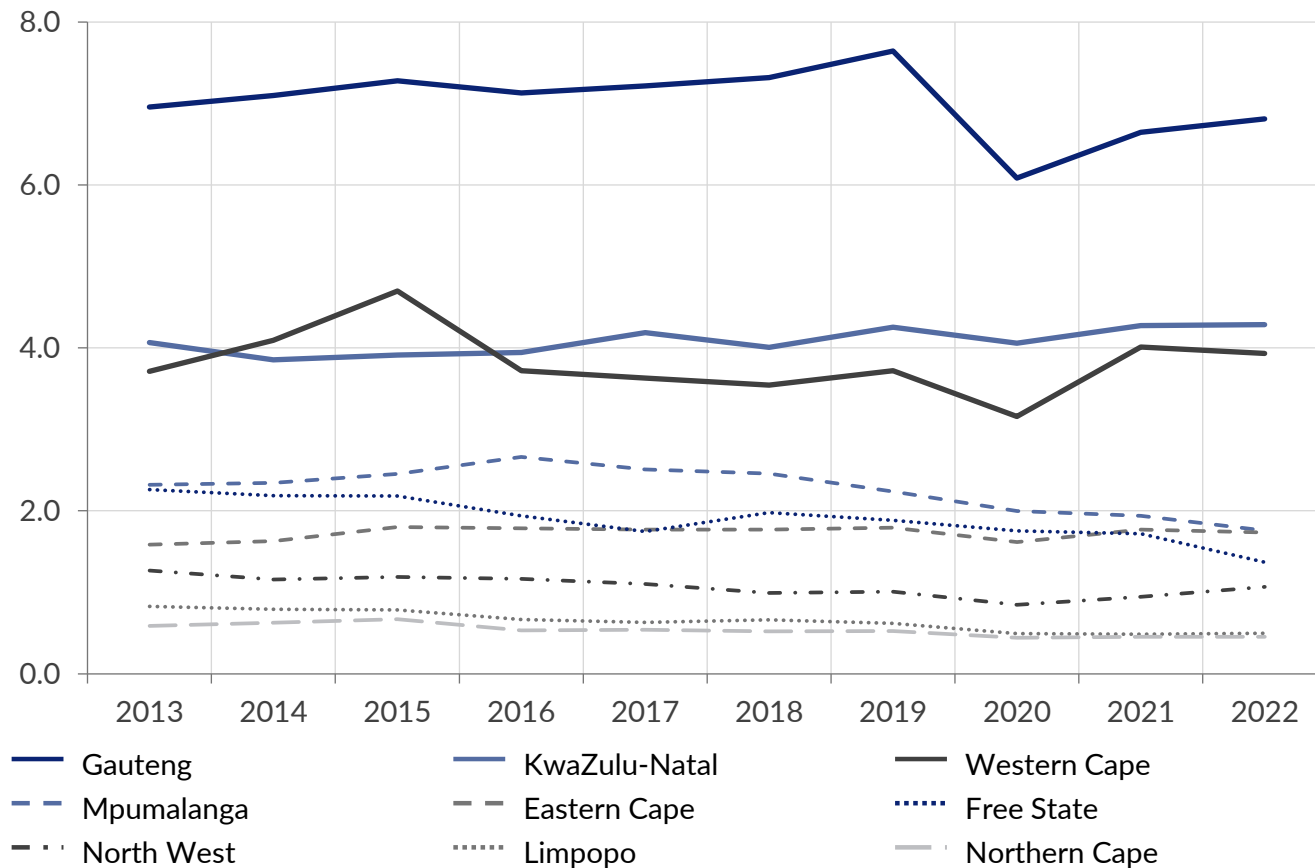
Colours show provincial sales totals, not density or customer catchments. Only Durban and Lesedi are labelled; competitor detail is on pages 13 and 15.

To resolve next

Nigel: confirm customer markets for Durban and Lesedi. Manish: retain the observed 2022 map; present later estimates separately with national-source flags.

S2. Reported fuel sales were broadly steady through 2022, despite the pandemic dip

Annual provincial sales | 2013–2022, bn litres/year



What the evidence shows

01 | Gauteng remains the largest market

The reported series places Gauteng above the other provinces throughout 2013–2022. Volumes differ across the smaller markets.

02 | Later provinces are estimates

2024 petrol estimates: Gauteng 3.47, KZN 1.47, Western Cape 1.41 bn L. These hold 2022 shares; no later observed provincial sales were found.

03 | Historical differences are traced

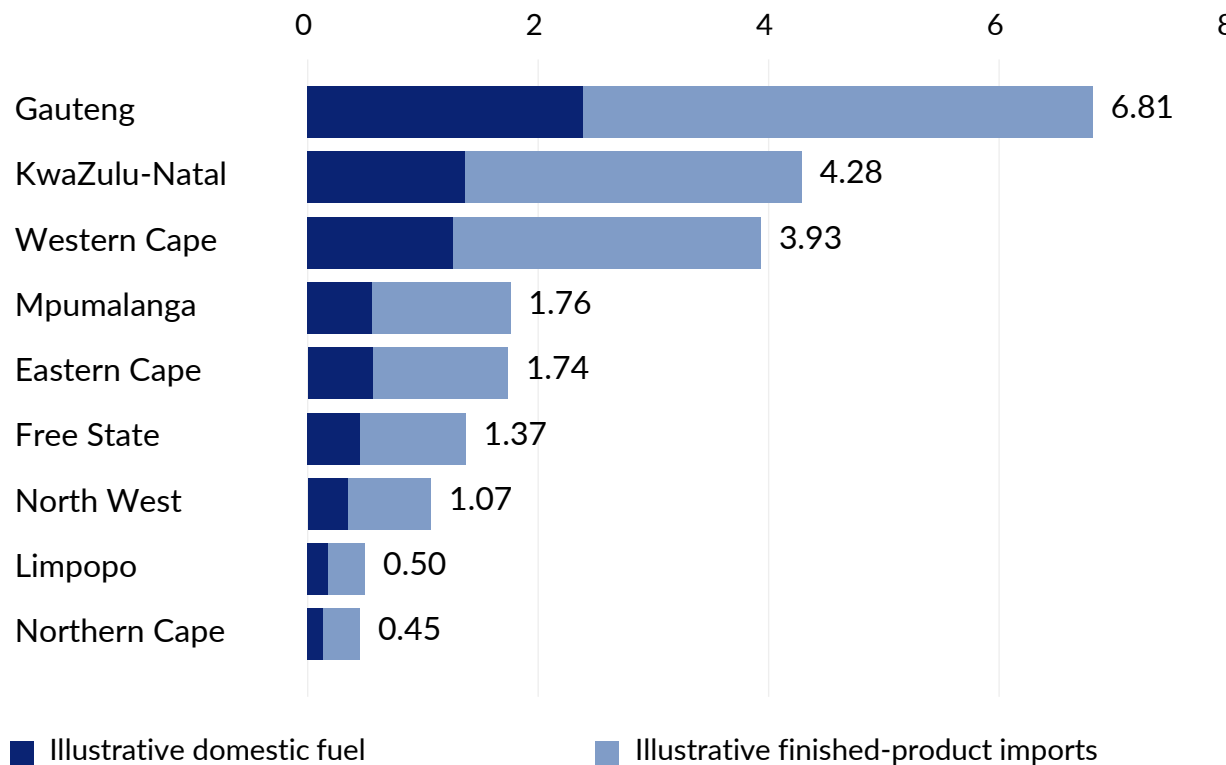
Manish fixed the 2013 parser and traced 2015 to a source revision. Decisions remain on 2014 conflicts and incomplete 2018 coverage.

To resolve next

Manish: reconcile source selections and retain estimate labels. Nigel: decide whether estimated petrol volumes belong in the outlook; diesel booking shares are less stable.

S1. Provincial demand is known; its supply origin still needs evidence

2022 sales totals | ILLUSTRATIVE domestic/import allocation



Units: bn litres/year. Imported/domestic segments are a scenario illustration, not measured provincial flows.

Source: DMPR 2022 provincial sales totals; source mix from the existing illustrative national-balance CSV. Not observed origin.

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What the evidence shows

01 | Demand totals are sourced

The provincial bars use reported 2022 petrol/diesel sales.

02 | Origin shares are illustrative

Domestic/import fractions are authored examples; no actual provincial origin allocation is established.

03 | Imports need a national balance

Exports and stock movements must be reconciled before a sourced import requirement can be presented.

To resolve next

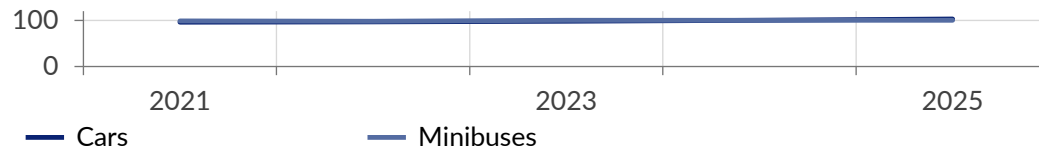
Manish: populate the matched-year national balance first; retain provincial origin as unknown until supported.

S/C3. Activity trends show potential demand shifts; fuel used per unit of activity still needs validation

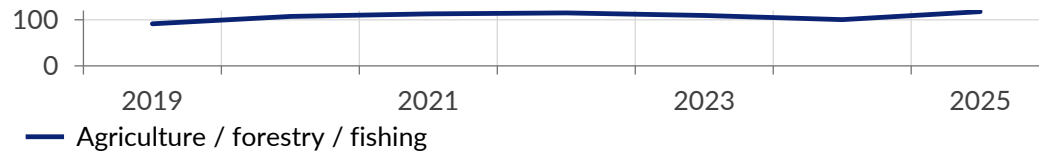
Potential demand levers | 2024 = 100; price panel uses nominal R/litre

Observed proxies, not fuel-volume effects. Activity axes 0–120; EV 0–400; prices 0–30 R/L.

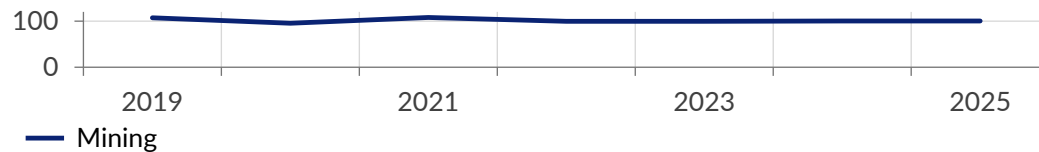
Passenger travel | all-fuel stock, Dec



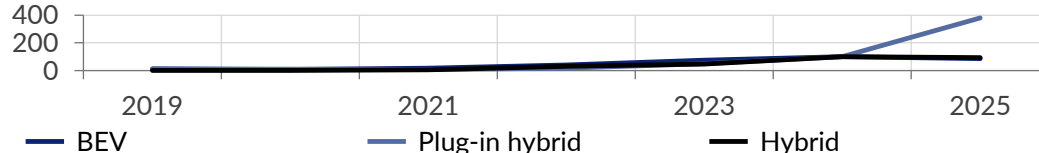
Agriculture | real value added, CY



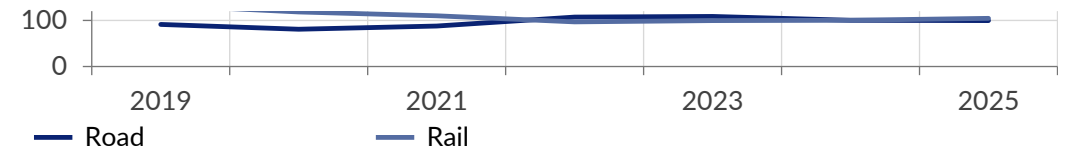
Mining | production volume, CY



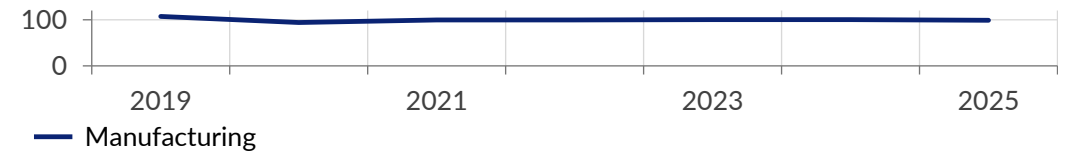
Electrification | annual new sales, CY



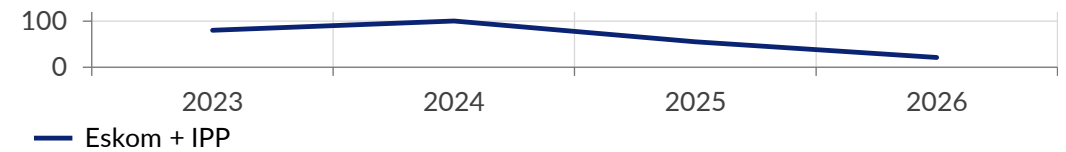
Freight demand | payload, CY



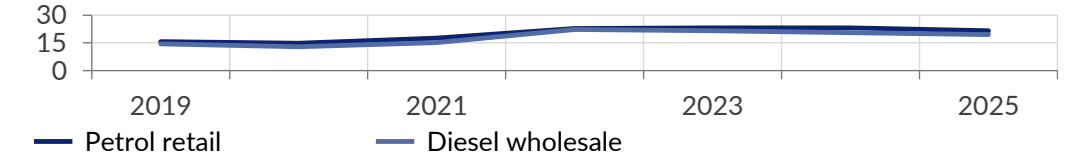
Manufacturing | production volume, CY



Diesel power | OCGT generation, FY



Price response | annual nominal R/litre



Note: Latest staged: activity Jul 2026; GDP Q2 2026; inland prices Oct 2026. Charts use complete years. Fuel-use rates need validation; FY ends March.

Source: NaTIS; Stats SA P0441 / P2041 / P3041.2 / P7162; Eskom; naamsa; DMPR / CEF. Snapshot manifest in notes.

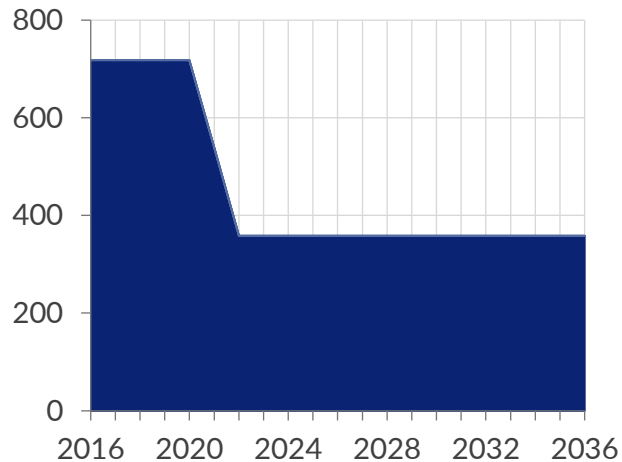
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C4. Refinery capacity has fallen; redevelopment remains conditional

Published footprint + illustrative scenarios | thousand bbl/day

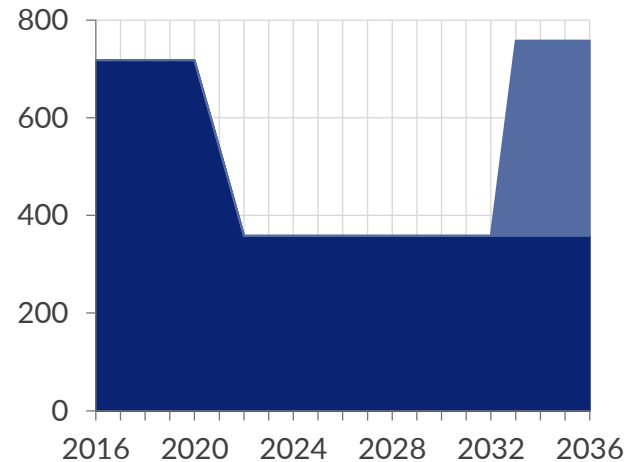
History 2016–2025 is sourced; 2026 onward is illustrative, not a forecast

A | No new refinery capacity



2036: 358 thousand bbl/day

B | Conditional CEF redevelopment



2036: 758 thousand bbl/day

Reported FY2024 output: Secunda 29.1; Natref Sasol share 17.8 million barrels.

What the evidence shows

01 | Capacity has fallen

Published capacity falls from 718 to 358 thousand bbl/day between 2020 and 2022. Redevelopment remains conditional.

02 | Reported output is different

FY2024 Secunda output is 29.1 million barrels; Natref is 17.8 for Sasol's share. Both cover all products and a year to June.

03 | A product bridge is needed

Match product yields, calendar years and downtime before calculating import effects. Capacity does not measure actual output.

To resolve next

Manish: match operator output to product/year and document Natref ownership basis. Henry: review technical mechanisms. Nigel: agree refinery scenario timing.

S/C3 / C4. L/M/H assumptions define coherent demand, supply and customer-access worlds

Scenario design | all key levers and assumptions

Input family	L/M/H settings to agree	Outputs and decision
Demand	Growth, mileage, fleet efficiency/EVs, freight/passenger rail, grid/private generation, sector activity and price response	Petrol/diesel demand by year and region
Supply and trade	Plant output, yields, downtime/compliance, imports, neighbouring-country exports and stocks	Domestic supply, trade and transit throughput; residuals explicit
Customer access	Delivered price, service, feasible routes, rights and switching	Reachable and competitively accessible customer litres
Capacity and operations	Usable tanks, receipt/dispatch, inventory policy, peaks and spare space	Unique Vopak flows and any usable-capacity gap

Combine settings into coherent worlds

Baseline	Ample supply	Tight supply
Agreed reference assumptions	Supply and access comfortably meet demand	Demand tests available supply and delivery routes

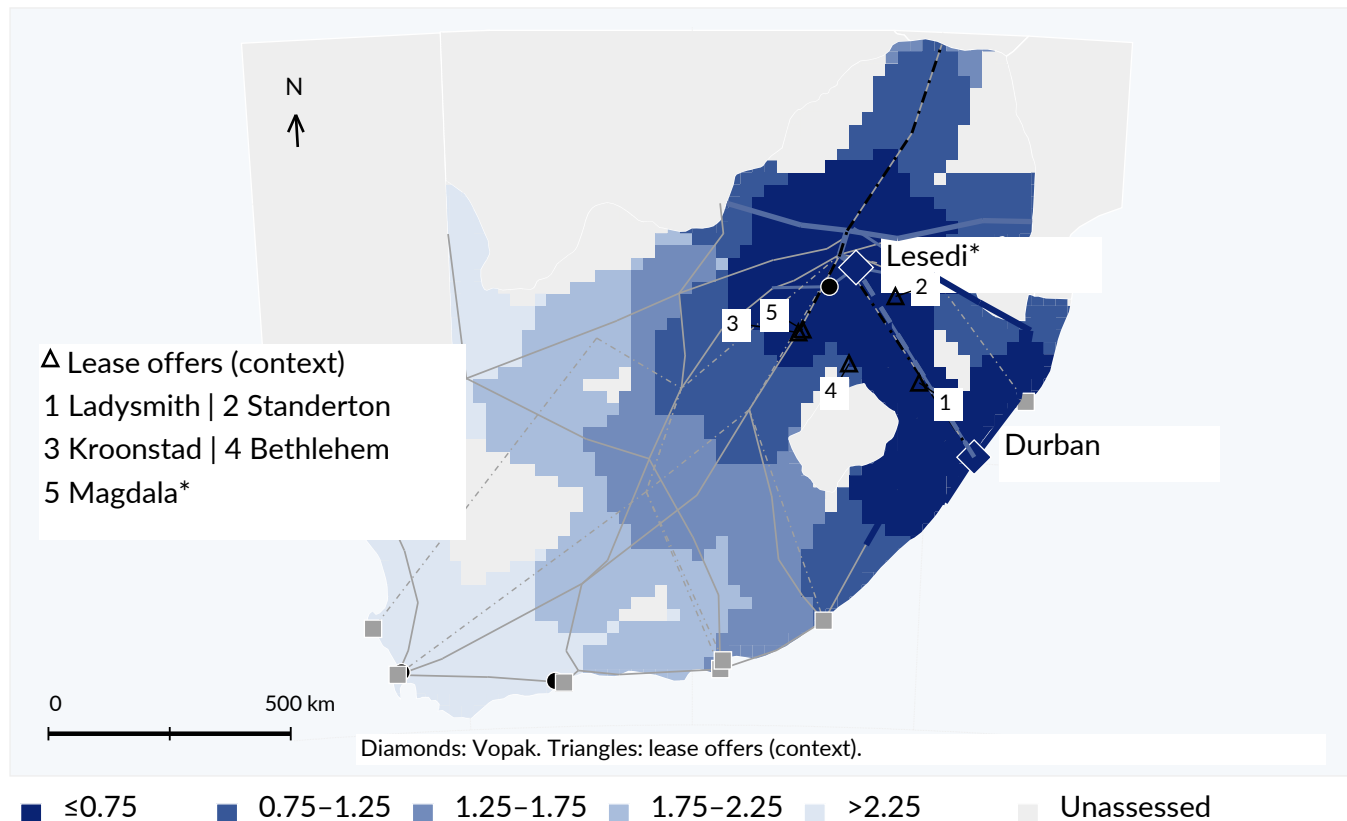
Note: History is observed; forecast cases are proposed. No new fuel forecast is shown until the assumptions, source selections and back-tests are reviewed.

Source: Authored forecast framework; observed series in reporting evidence snapshots. No new growth rates or fuel forecasts inferred.

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C5. Transport cost shapes the market Durban and Lesedi can reach

Illustrative road cost, R/litre



Example rate: R0.15/L dispatch + R0.002/L/km. Grey = unassessed.

What the evidence shows

01 | Darker means lower transport cost

Cells show example road cost from Durban or Lesedi. The rate is illustrative and excludes fuel price, storage and pipeline tariffs.

02 | Grey means unassessed

Grey areas have no assessed road connection under this illustration. They are not confirmed to be inaccessible.

03 | A cost surface is a first screen

Pipeline and rail lines are context. Route capacity, product compatibility and customer rights still determine serviceable demand.

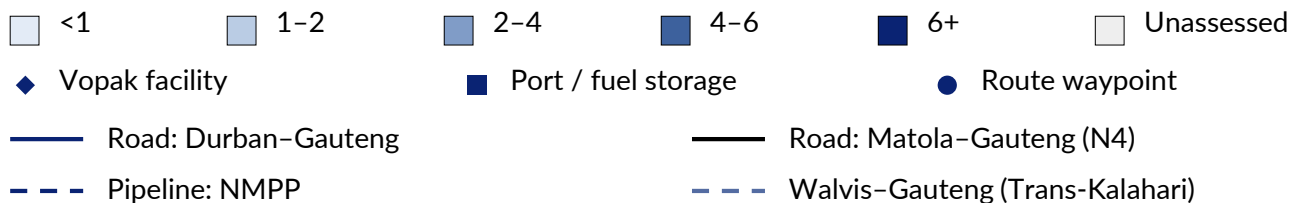
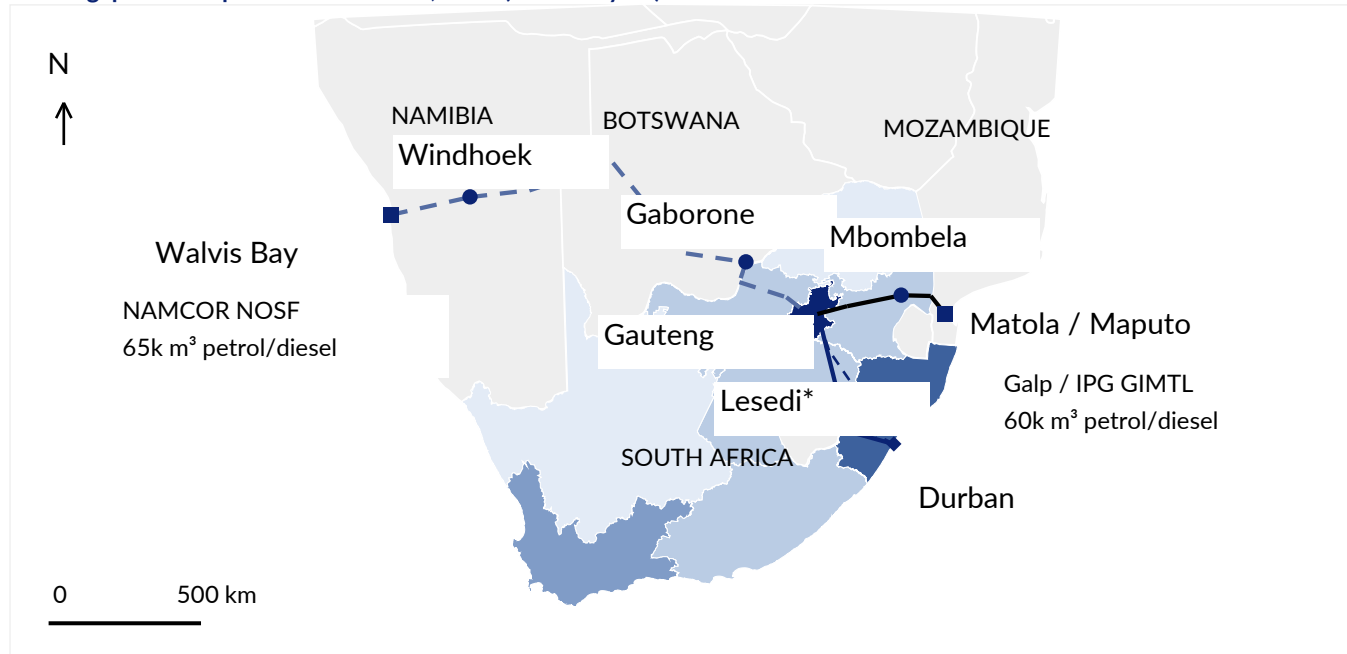
To resolve next

Nigel: select the first customer destination and cost threshold. Manish: replace example rates with dated route costs and access evidence.

C5. Compare Durban, Matola and Walvis Bay for the same inland customer

Durban, Matola and Walvis Bay | schematic corridor comparison

Shading: provincial petrol + diesel sales, 2022 (bn litres/year)



Note: Routes schematic; access unverified. Colour measures provincial totals, not density or Vopak share.
 Source: DMPR 2022 sales; NAMCOR NOSF; Galp GIMTL; WBCG / TRAC corridors; Natural Earth. Capacities and URLs in notes.
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What the evidence shows

01 | Vopak has a South African footprint

Durban and Lesedi are highlighted against 2022 provincial sales. Colour shows sales totals; it does not establish Vopak market dominance.

02 | Matola has published fuel storage

Galp/IPG GIMTL lists 60,000 m³ of petrol/diesel tanks, plus 6,000 m³ LPG excluded here. N4 is the eastern inland comparison route.

03 | Walvis has storage and handling

NAMCOR NOSF lists 75,000 m³ in total, including 65,000 m³ petrol/diesel. Trans-Kalahari is the western inland comparator.

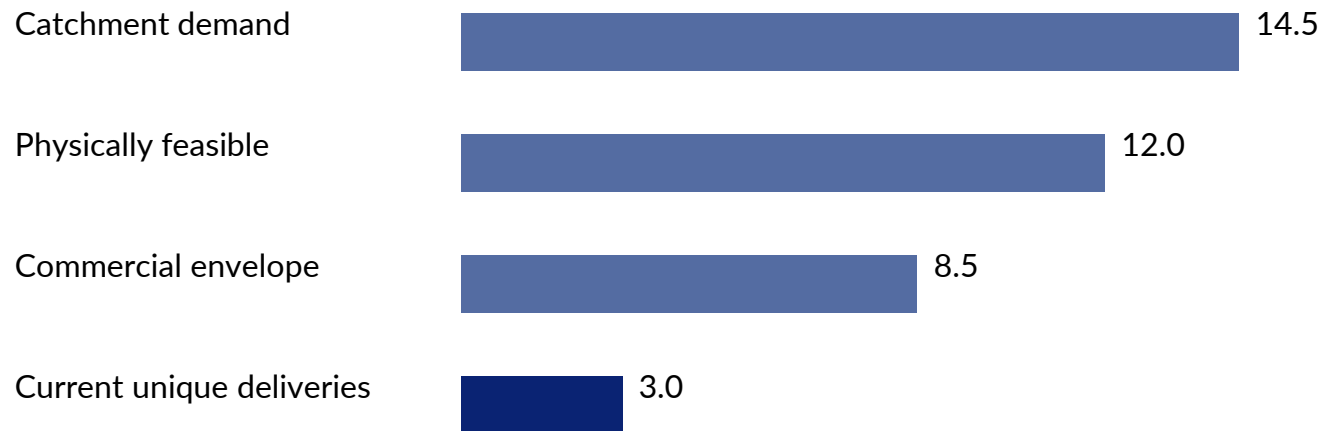
To resolve next

Manish: verify available tanks, customer access and full delivered costs. Source Namibia/Mozambique demand before extending the heatmap; grey means unassessed.

R6. Separate market demand, accessible volume and unique Vopak deliveries

Market screens | illustrative bn litres/year

Successive screens; these volumes are illustrative



Count Durban-Lesedi transfers once

Durban 2.8 + Lesedi 2.0 – shared transfer 1.8 = unique 3.0

Demand ≠ terminal receipts ≠ storage capacity. Incremental opportunity is on page 14.

What the evidence shows

01 | Count customer demand once

Example receipts of 2.8 at Durban and 2.0 at Lesedi include a shared 1.8 transfer. Unique demand served is therefore 3.0 bn litres.

02 | Accessibility precedes opportunity sizing

Example catchment demand 14.5 → physically feasible 12.0 → commercial envelope 8.5. Each screen requires evidence.

03 | Examples are separate from sales

These market volumes illustrate the calculation. They are not the reported 2022 regional totals or actual Vopak deliveries.

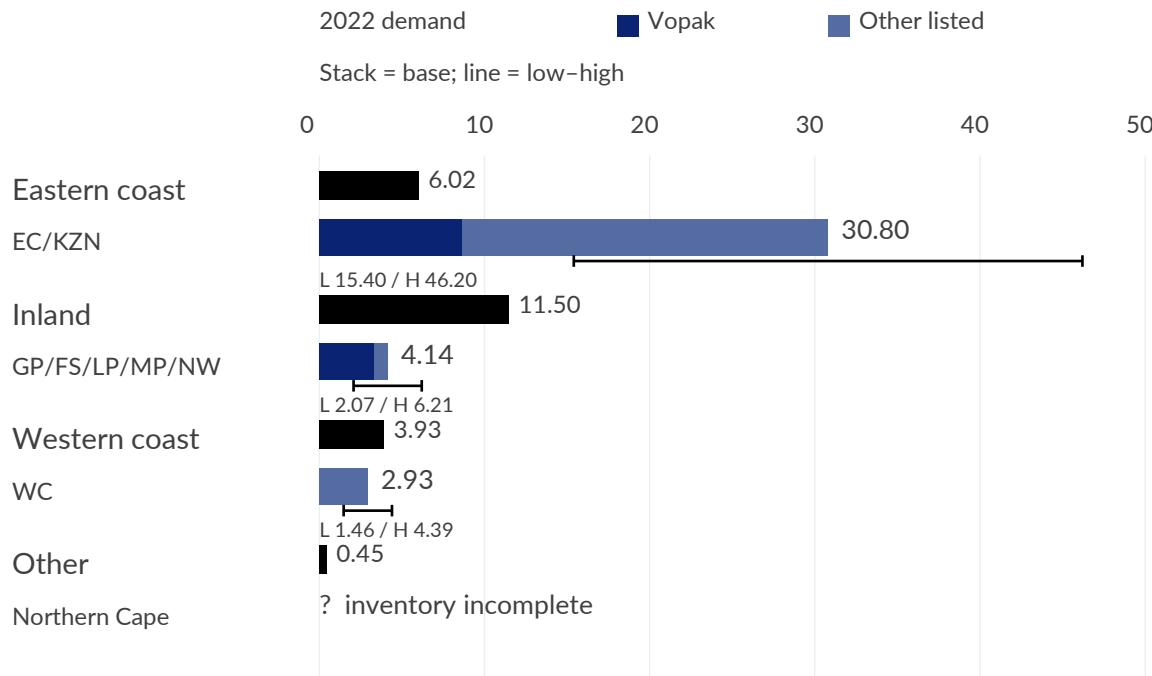
To resolve next

Nigel: request customer deliveries, destinations and transfer records. Manish: calculate unique served demand before estimating additional access.

R6. Tank turnover turns the storage footprint into an annual handling estimate

Annual demand and gross handling equivalent | million m³/year

Provisional turns/month: low 1 / base 2 / high 3



Same annual scale. Gross tankage × monthly turns × 12; not available fuel supply.

Historical 2022 sales vs inventory checked 2026. Mixed products; missing sites; lease offers excluded. No regional shortage or market share is established.

[1] Vopak [2] Bidvest [3] Burgan [4] Shell [5] NERSA [6] Sasol [7] Transnet

Source: DMPR 2022 sales; published site inventory; Blackmer/Dover 2013; Wood Mackenzie 2009. Turnover cases authored.

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What the estimate means

01 | Estimate now; replace with site data

Use 1 / 2 / 3 monthly turns to test the range. Two is an authored base, informed by generic industry material; no SA site rate is verified.

02 | Operating limits set achievable volume

Gross handling equivalents assume the listed tankage can turn. Fuel compatibility, working space, downtime and receipt/dispatch limits still need confirmation.

03 | Missing tankage is not a shortage

Inland coverage omits material operator capacities. Coastal handling can also serve inland customers. Do not add regional handling as unique national sales.

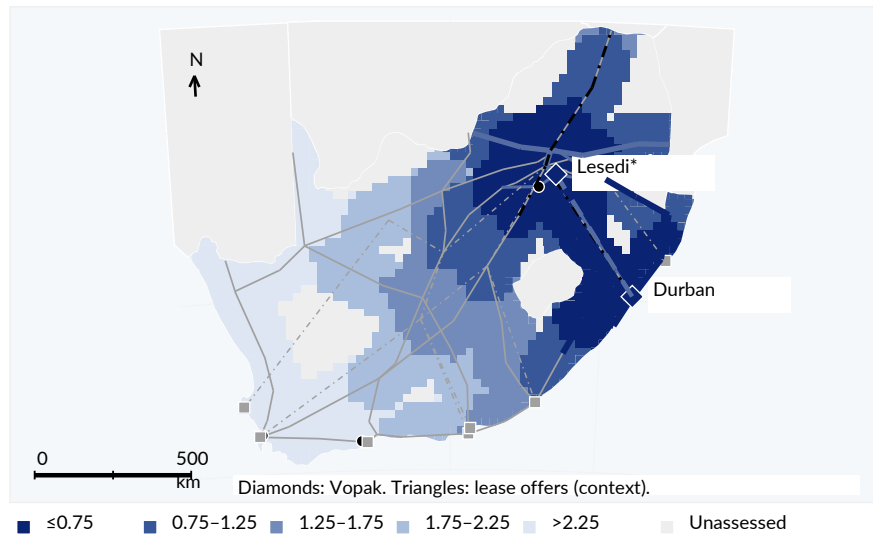
Next actions

Manish: verify dispatch and working capacity at the p15 sites.
Nigel: obtain client operating data. Replace the common range with site-specific turns.

R6. Reach and commercial access narrow the illustrative customer opportunity

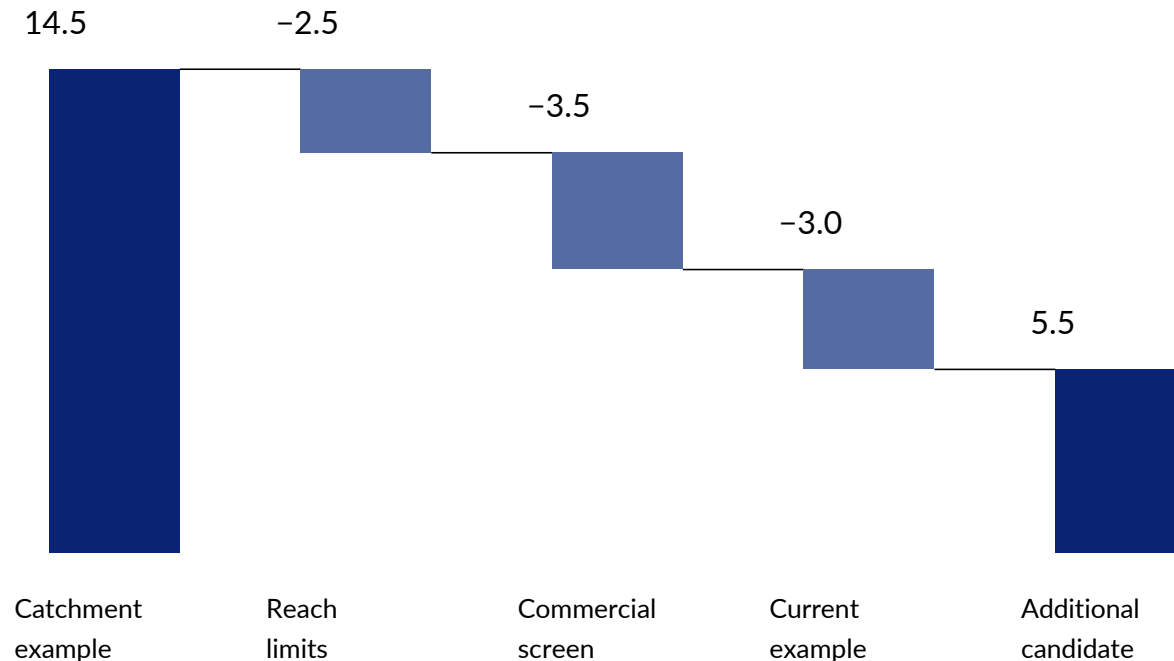
Illustrative reach and volume bridge | bn litres/year; customer capture unverified

Durban and Lesedi: illustrative road reach



Darker shading = lower example road cost. Physical and commercial access remain unverified.

Illustrative volume waterfall

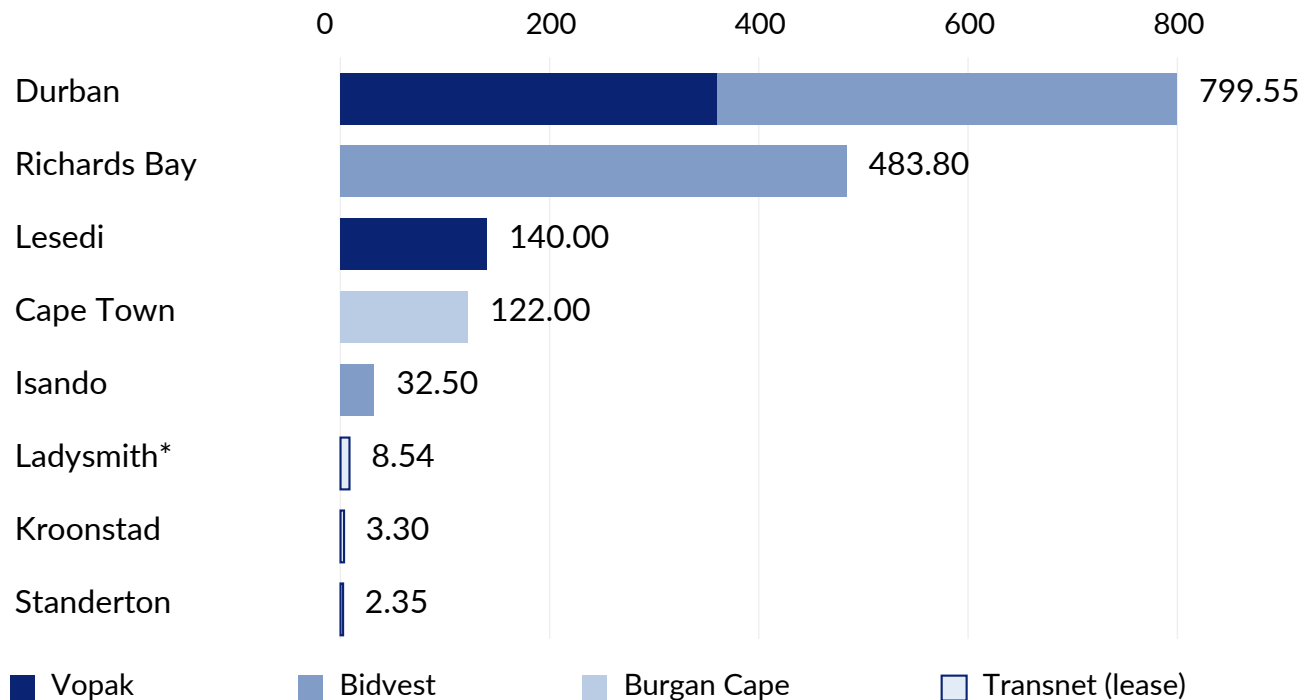


The example leaves 5.5 bn L/year of candidates to test for cost, service, usable capacity and customer rights.

R7. Durban leads the published storage subset

Published storage by location | thousand m³; mixed capacity bases

One shared scale for every location



Small lease bars reflect smaller tank volumes; they are not operating supply.

What the evidence shows

01 | Turnover connects this stock to p13

At 1 / 2 / 3 turns/month, p13 estimates annual handling from this inventory. Actual deliveries and customer share still need evidence.

02 | The inventory is partial

Missing capacities are unknown, rather than zero; mixed product scopes limit comparison.

03 | Leases are conditional

Transnet lease tanks are not counted as currently available operating supply.

To resolve next

Manish: verify product compatibility, capacity basis and operating status. Nigel: confirm usable client capacity.

R7. Prioritise customer markets, win on service and cost, then expand

Market outlook | priority hypotheses and conditional timing

			
Where to play	How to win	When to act	What must be true
Durban coastal customers First market to test	Competitive delivered cost; reliable receipt and dispatch; compatible tanks	Now: identify customers and test service using existing assets	Named customer, product and unique litres; cost threshold; usable tank and dispatch space
Lesedi inland customers First inland market to test	Reliable inland supply; competitive full-route cost versus other gateways	After route and customer validation: negotiate incremental flows	NMPP or alternative route access; receipt/dispatch capacity; customer rights and switching
Matola / Walvis Bay port-storage gateways Compare inland supply routes	Test storage, handling and transport to the same inland destination	During the inland comparison; pursue an opportunity only if access and economics are evidenced	Available compatible tanks; border/route feasibility; partner and customer access; full delivered cost

Investment gate

New storage follows secured incremental flows and a gap in usable capacity; the illustrative 5.5 bn litres does not establish a commitment.

Nigel: Choose the first customer/product and acceptable cost. Confirm the service proposition and commercial access.

Manish: Build a customer-by-route evidence table: litres, cost, capacity, rights, timing and owner. Mark each gate passed, failed or unassessed.

Source: Authored market hypotheses; demand, gateway and storage evidence on pp.4, 10–15. Customer capture remains unverified.

Appendix

The main story ends on page 16.

Document scope · Henry storyboard responses · supporting presentation palette

Use the supporting material to trace the evidence, assumptions and remaining work behind the outlook.

About this document

What it covers

South Africa petrol/diesel market baseline, historical demand and national supply/trade evidence.

Demand drivers, refinery alternatives and L/M/H scenario design for all key levers and assumptions.

Vopak customer opportunities through infrastructure reach, delivered price, service and commercial access.

Preliminary inventory sensitivities and a trace of the evidence needed to answer Henry's storyboard.

Week 1 working assessment, 6 October 2026. Observed data, estimates, conditional cases and illustrations are distinguished. Source gaps and proposed actions remain explicit; governance coverage is not data validation or business approval.

What it does not establish

A validated 2026 forecast or calibrated L/M/H scenario results; assumption ranges and back-tests remain work to do.

Measured Vopak market share, confirmed customer capture or verified operational catchments.

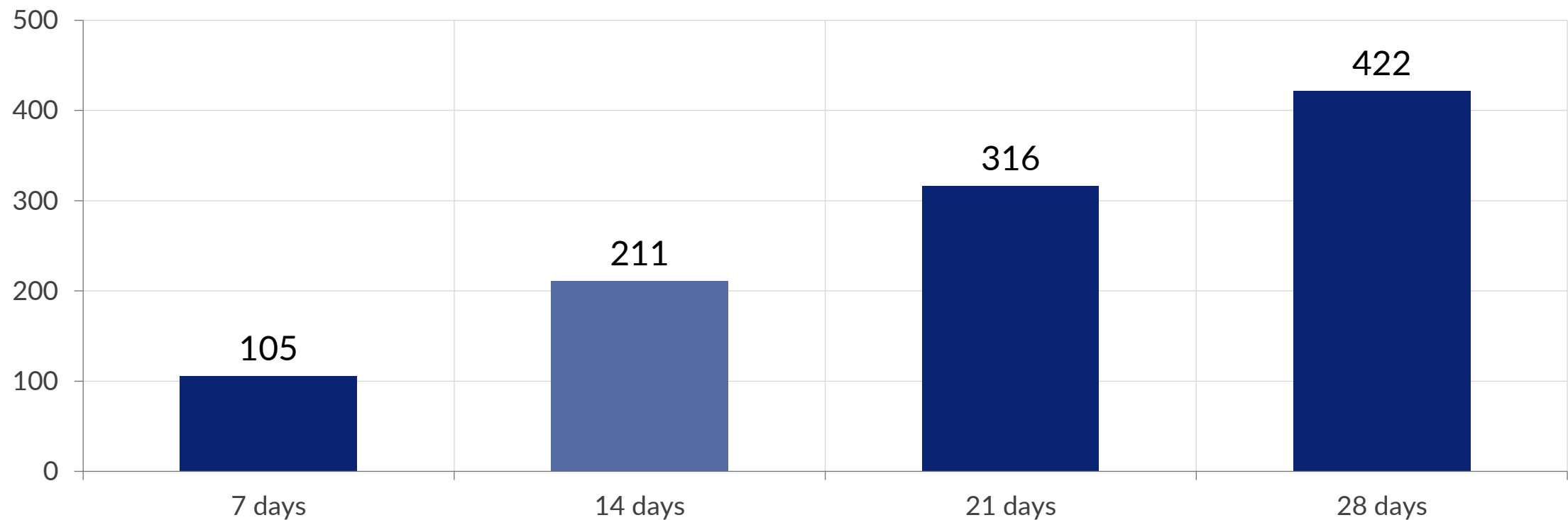
A closed national fuel balance, complete fuel-compatible storage census or confirmed spare tank capacity.

An investment recommendation, detailed engineering design or approved commercial business case.

R7. Inventory policy determines the working stock needed for additional flows

Working inventory, thousand m³

Illustrative: 5.5 bn litres/year; uniform daily flow



Inventory days

Working stock, not new capacity: allow for spare tanks, peaks, product segregation, tank heels and dispatch limits.

Source: Existing 5.5 bn L/year candidate example; authored inventory-day sensitivity, 6 Oct 2026. Not new-capacity sizing.

S1 / S2 / S/C3 / C4. Henry storyboard answers | Market baseline

S | Every prompt addressed; partial and open findings require the stated evidence

Henry storyboard prompt	Current answer / evidence gap	Deep dives
Total SA demand development	Partial: reported 2024 sales are 20.763 bn L; coverage awaits reconciliation. Provincial history ends in 2022.	3 / 4 / 5
Eastern SA market within economic striking distance	Open: location and illustrative costs show reach, not accessible customer litres. Need destinations, tariffs and rights.	4 / 10 / 12 / 14
Greater diesel use in power generation	Partial: generation history is shown; recent OCGT use falls. Diesel-litres attribution still needs fuel-use evidence.	7 / 9
Rail to road	Partial: road/rail payload history is shown. Payload alone does not prove modal substitution or diesel effects.	7 / 9
Changes in fuel price	Partial: complete-year price history is staged. Travel response and real-price effects remain uncalibrated.	7 / 9
More fuel-efficient vehicles	Open: fleet stocks and new EV/hybrid sales do not establish fleet efficiency. Need fuel mix, mileage and intensity.	7 / 9
Slow GDP growth	Partial: GDP and sector activity evidence is available; growth-to-fuel effects require explicit assumptions.	7 / 9
Refinery closures	Partial: published capacity falls to 358 thousand bbl/day. Capacity is distinct from output and operating availability.	8
Challenges at Secunda	Partial: FY2024 output is 29.1 million barrels, all products. Availability and petrol/diesel yields remain open.	8 / 9
Closure of Moss gas	Open: plant-specific shutdown dates, operating status and lost product output need explicit source confirmation.	8 / 9
Growth in imports and where they enter	Partial: 2024 net imports are 13.078 bn L. Import-growth attribution, product-specific ports and origin remain open.	3 / 6

Source: Henry storyboard pp30–32; reporting evidence and explicit gaps, 6 Oct 2026. Responses do not establish closure.

S/C3 / C4 / C5. Henry storyboard answers | Market changes

C | Every prompt addressed; partial and open findings require the stated evidence

Henry storyboard prompt	Current answer / evidence gap	Deep dives
Road to rail; private-sector rail projection	Open: historical payload is available; private-sector service, timing and modal-shift assumptions need evidence.	7 / 9
End of loadshedding and spare power capacity	Partial: observed OCGT generation falls. Future dispatch, spare capacity and diesel demand remain conditional.	7 / 9
EV / HEV penetration projection	Open: sales uptake is shown, not fleet penetration. Separate BEV, PHEV and conventional hybrids; model turnover.	7 / 9
Effect of EV / HEV on litres per km	Open: require stock mix, mileage and fuel intensity. BEVs and hybrids have different fuel-use mechanisms.	7 / 9
Higher fuel prices and km travelled	Open: prices are sourced; mileage response and elasticity must be specified and tested before forecasting.	7 / 9
Mega refinery / SAPREF restart	Conditional: redevelopment capacity is a scenario, not sanctioned output. Confirm FID, ramp-up and product yields.	8 / 9
Secunda gas, coal and MRG diversion to 2032	Open: no liquid-fuel displacement is inferred. Need dated gas allocation, plant mechanism and product-output evidence.	8 / 9
Natref future?	Open: reported FY output is available; future ownership, availability and product yields require a dated case.	8 / 9
Durban storage / NMPP / Lesedi vs road and rail	Open: compare the same customer and product. Need full delivered costs, capacity, tariffs and service availability.	10 / 11 / 14
Transport from Matola and Walvis Bay	Partial: gateways, published storage and schematic routes are shown. Full costs, borders and customer access remain open.	11 / 14 / 16

Source: Henry storyboard pp30–32; reporting evidence and explicit gaps, 6 Oct 2026. Responses do not establish closure.

R6 / R7. Henry storyboard answers | Vopak outlook

R | Every prompt addressed; partial and open findings require the stated evidence

Henry storyboard prompt	Current answer / evidence gap	Deep dives
Market share of Vopak in Dbn and Lesedi	Open: published storage is an asset footprint, not served demand. Need unique deliveries, transfers and matched demand.	12 / 13 / 14
Potential scope for new investment?	Conditional: prioritise customer flows and service first. New storage requires secured incremental flows and a usable-capacity gap.	15 / 16 / 19

Agree priorities and next steps

Confirm demand, routes and access

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Lesedi | vopak.com



Presentation palette | colours and HEX codes

Reference | colours used in this presentation

Core presentation colours

	Vopak blue #0A2373		Secondary blue #546CA2		Text / ink #000000		White #FFFFFF
	Unassessed / neutral #EEEEEE		Header divider #A0A0A0		Body divider #B8B8B8		Frame divider #B0B0B0

Sales-map colours | provincial totals, bn litres/year

	Below 1 #E3EBF6		1 to below 2 #BACCE4		2 to below 4 #809CC7		4 to below 6 #3D619D
	6 and above #0A2373		Map background #F5F8FB				

Other template theme colours

	Theme dark 2 #11151A		Theme light 2 #E7E6E6		Accent 3 #BDBEC1		Accent 4 #404040
	Accent 5 #767676		Accent 6 #D9D9D9				